

ORFEUS seen from its User Advisory Group

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2023 EPOS Seismology Workshop, 9 – 11 October, Podgorica, Montenegro



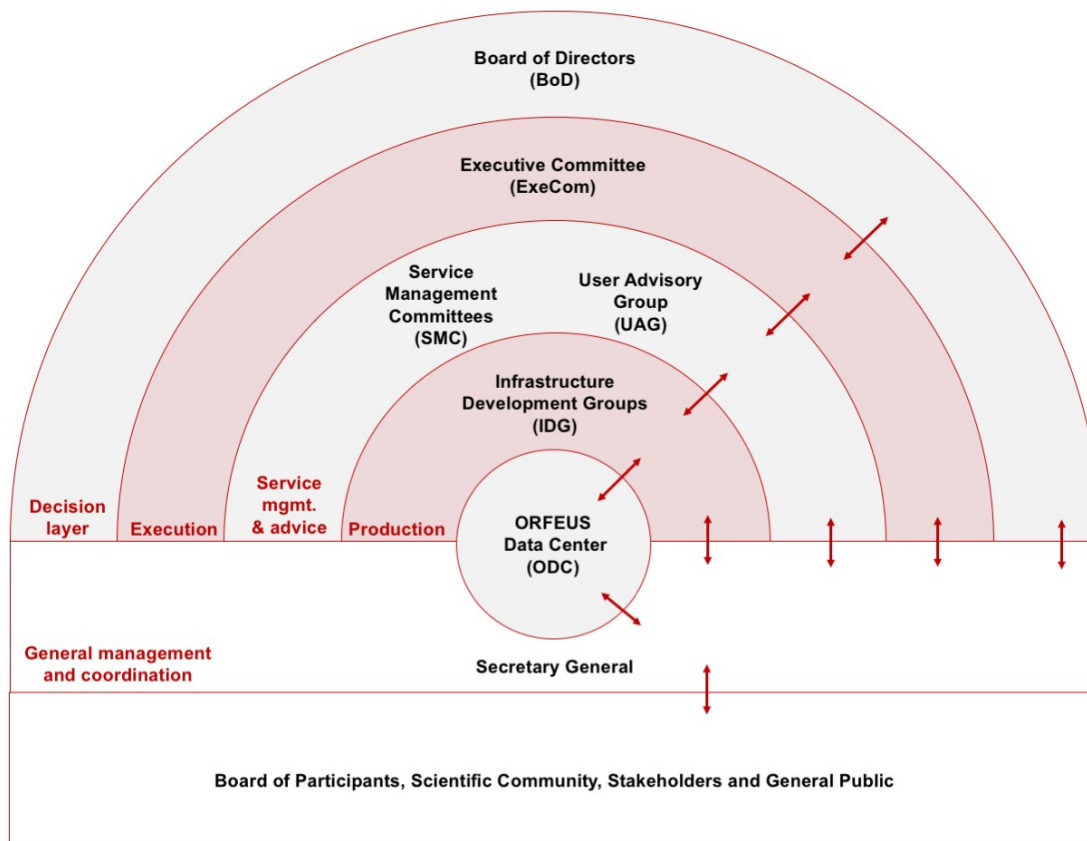
INSTITUTE OF GEOPHYSICS
OF THE CZECH ACADEMY OF SCIENCES



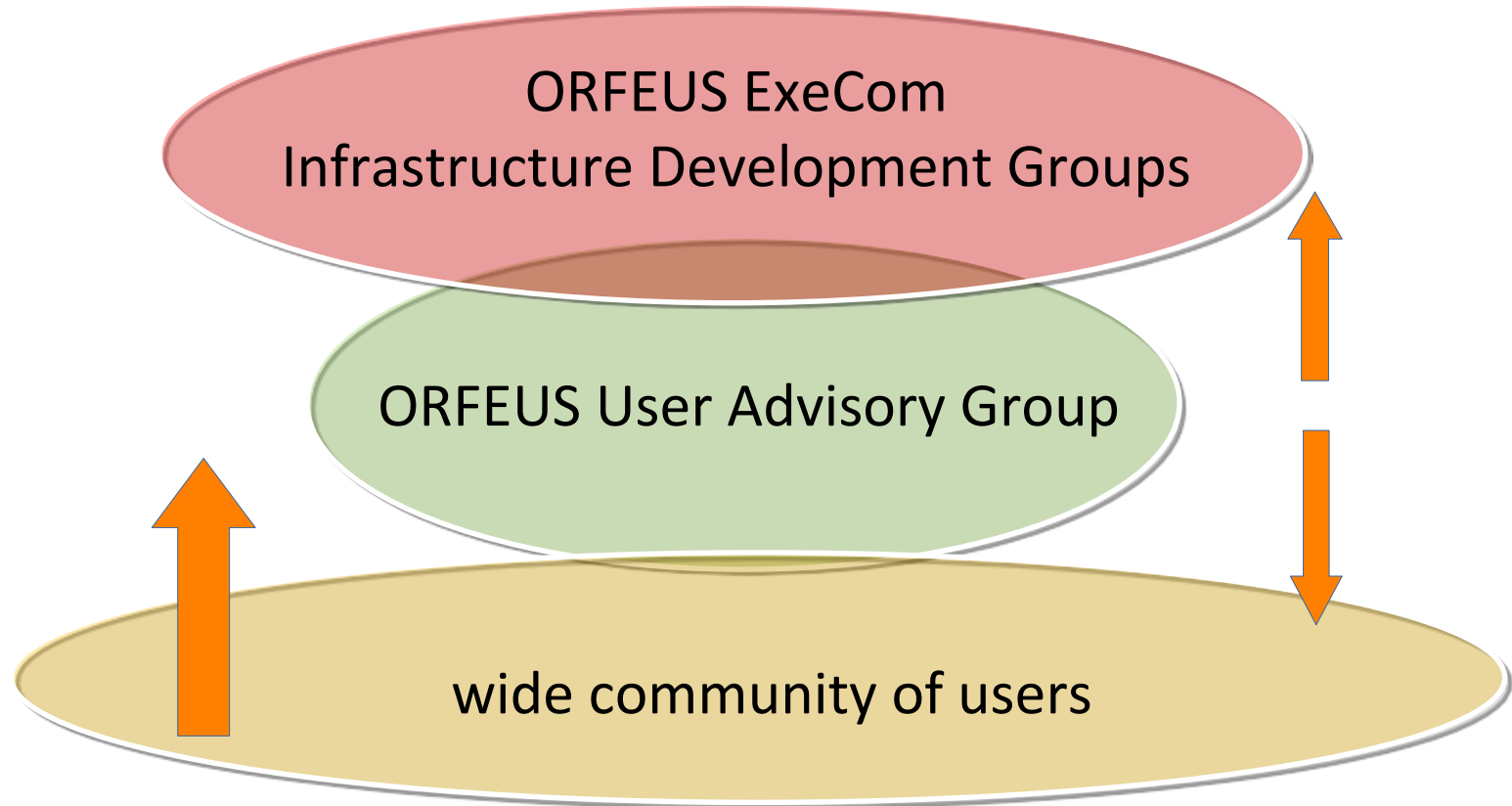
today's agenda

- what is UAG
- history of UAG
- reports, particularly the 2023 report to ORFEUS
- user community input
- short-term vision
- vision for next years

ORFEUS governance – where is UAG?



what is UAG? (1)



what is UAG? (2)

From ORFEUS Bylaws:

Section 3. User Advisory Groups (UAGs). The BoD shall establish at least one User Advisory Group (UAG) to serve as an experienced advisory body for the ExeCom. A UAG is composed of at least four external stakeholders with relevant scientific or technical competence. The members of UAGs are proposed by the ExeCom and appointed by the BoD. At least one member of each UAG should come from an organization that is not a Core Participant of ORFEUS. Each UAG meets once a year with the SG, the President of the ExeCom and with the Chairs of the relevant SMCs, and subsequently produces a yearly report with scientific and technical recommendations. The term for members of the UAG(s) is 2 years, with renewal possible; the UAG(s) elects a Chair among its members every two years. Upon recommendation of the ExeCom, the outgoing Chair of the UAG may extend his/her UAG membership by one year to ensure effective continuity on any pending issues.

Section 4. Powers of the UAGs. UAGs review practice and services and suggest improvements to the ExeCom on all matters related to the implementation of the services and products. They advise the ExeCom on the prioritisation of new tasks and services.

Kick-off meeting 2018:

The role of the UAG is to ensure awareness of the ORFEUS ExeCom of the main scientific, present and future challenges in Seismology and Earth Sciences, and to provide feedback on how ORFEUS is serving its Community to face them.

history of the ORFEUS User Advisory Group

2018, July 30, updated ORFEUS governance:

The products and services delivered by ORFEUS will be planned, assessed and improved through the technical and scientific feedback of a newly established **User Advisory Group (UAG)**, comprised of active European Earth scientists with expertise encompassing a broad range of disciplines. The members of the UAG are: Thomas Meier (Chair), Pierre Boué, John Douglas, Florian Fuchs, František Gallovič, Lion Krischer, Olga-Joan Ktenidou, Stéphane Rondenay.

	member			chair			
	2018	2019	2020	2021	2022	2023	2024
Pierre Boué							
John Douglas							
Florian Fuchs							
Lion Krischer							
Thomas Meier							
František Gallovič							
Olga-Joan Ktenidou							
Stéphane Rondenay							
Dino Bindi							
Petr Kolínský							
Antje Schlömer							
Luděk Vecsey							
Felix Eckel							
Davor Stanko							
Valerio Poggi							

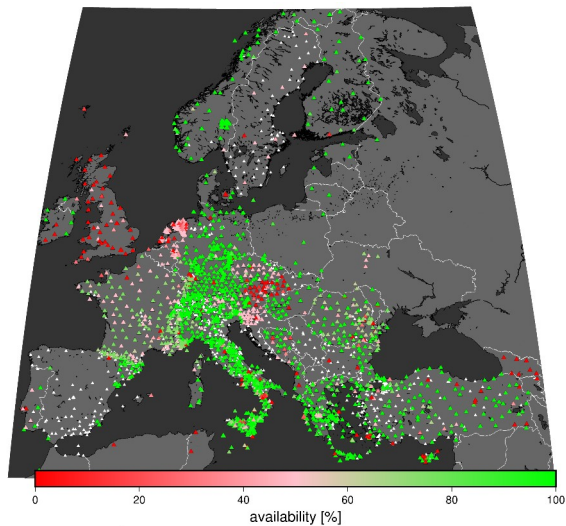
UAG reports to ORFEUS

1 st report	SM	November 2018	data access, portals
	BB	January 2019	data availability
2 nd report	SM	February 2020	follow-up on the previous
	BB	January 2020	data availability + quality
3 rd report	BB+SM	October 2021	(meta)data quality tools
4 th report	BB+SM	September 2023	metadata quality tools

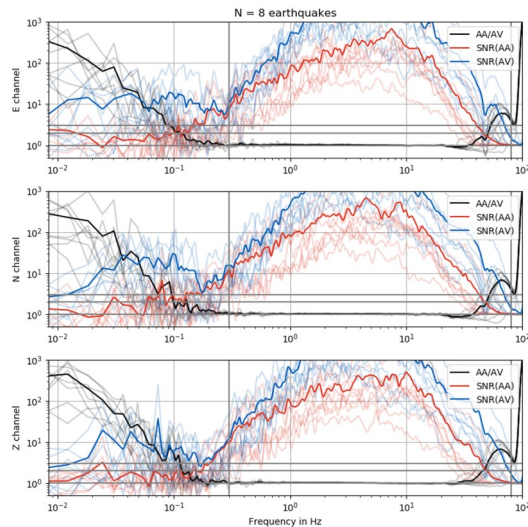
- submitted in September 2023
- 21 pages, 15 figures, 20 recommendations
- covers the activities of UAG in 2022 and 2023
- based on the:
 - work done by UAG members
 - work done by other interested users
 - replies by ORFEUS bodies to the previous report (end of 2021)
 - input from the community gathered during workshops, conferences, personal meetings, ...

UAG 2023 report – tests (1)

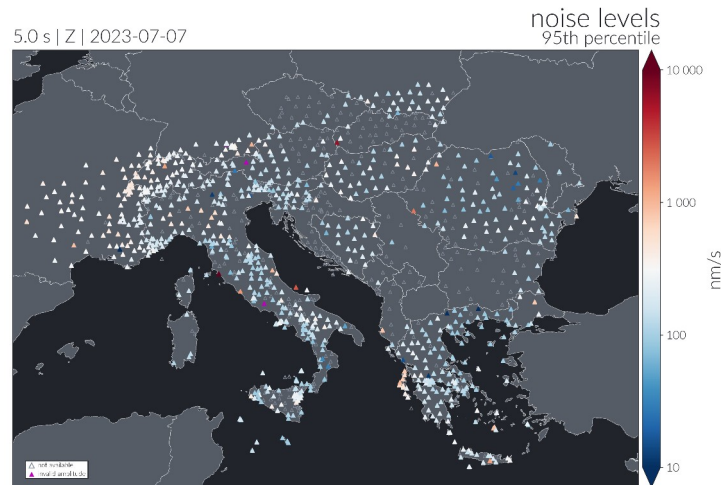
retrievability tests
by Johannes Stampa et al.



SM + BB comparison
by Carlo Cauzzi et al.

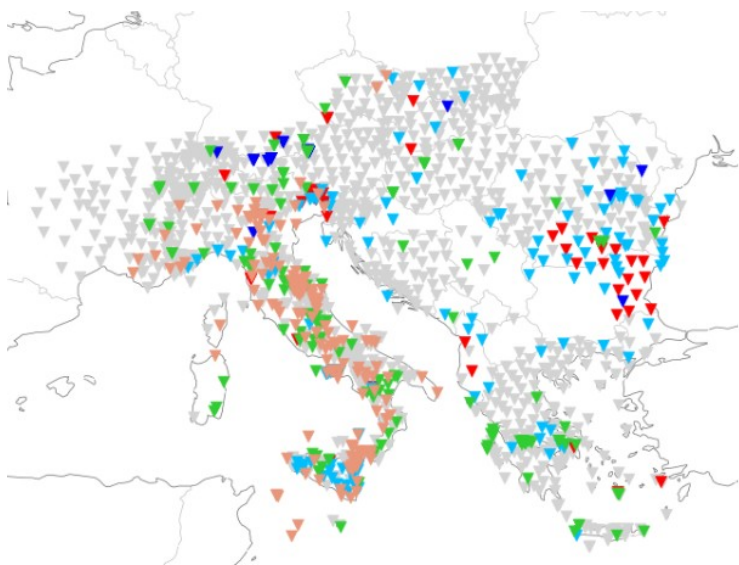


noise maps by Felix Eckel



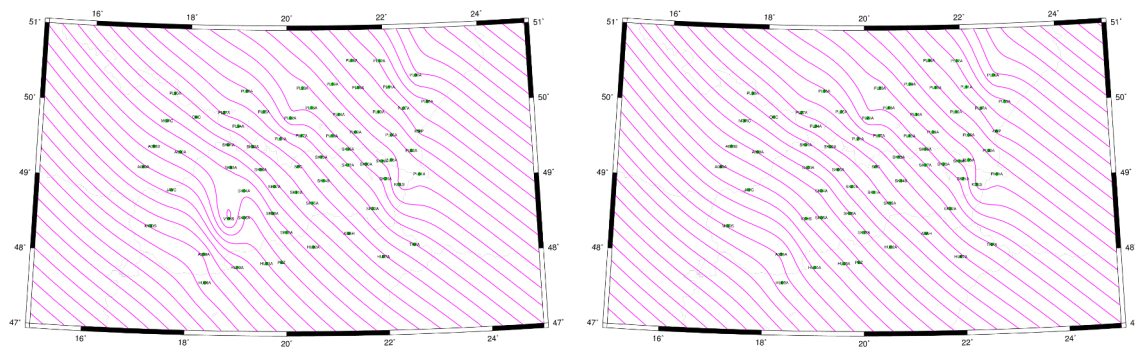
UAG 2023 report – tests (2)

formal StationXML checks
and corner period cross-check
by Luděk Vecsey



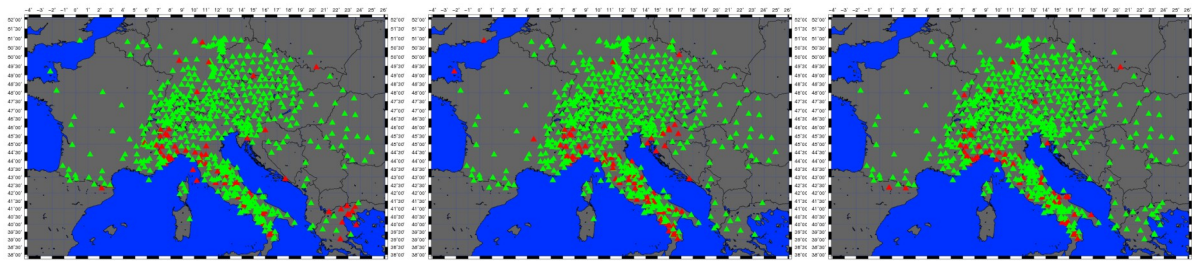
- orange triangle metadata warnings (15.3 %)
- red triangle errors in metadata (3.6 %)
- green triangle wrong band code (8.4 %)
- blue triangle different corner periods in metadata and the list (8.8 %)
- purple triangle switched broadband and short-period (1.0 %)

total: 37 %



wavefront test for detecting
phase characteristics issues
by Petr Kolínský

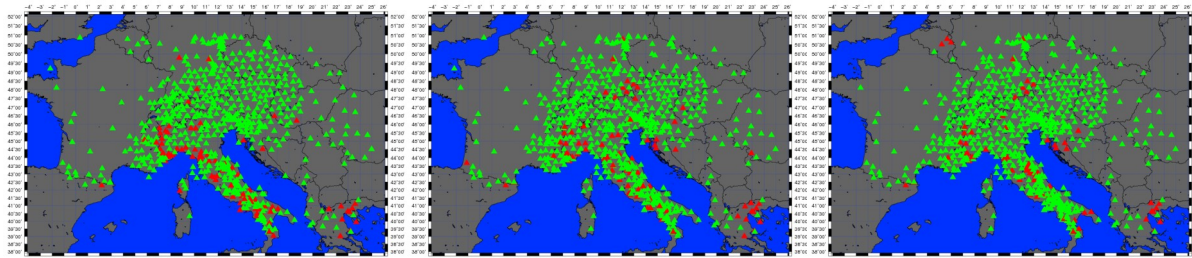
UAG 2023 report – data Q - look back



Kamchatka M=7.2, 30 **January** 2016

Sumatra M=7.8, 2 **March** 2016

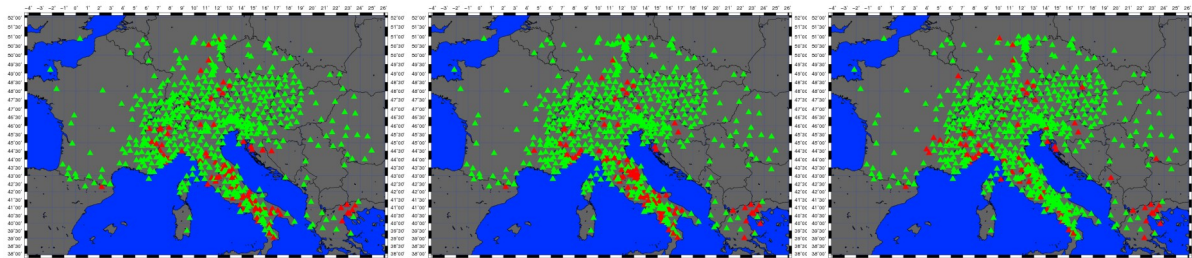
Morocco M=5.2, 15 **March** 2016



Ecuador M=7.8, 16 **April** 2016

South Atlantic M=7.4, 19 **August** 2016

Ascension M=7.1, 29 **August** 2016



Greece M=5.5, 15 **October** 2016

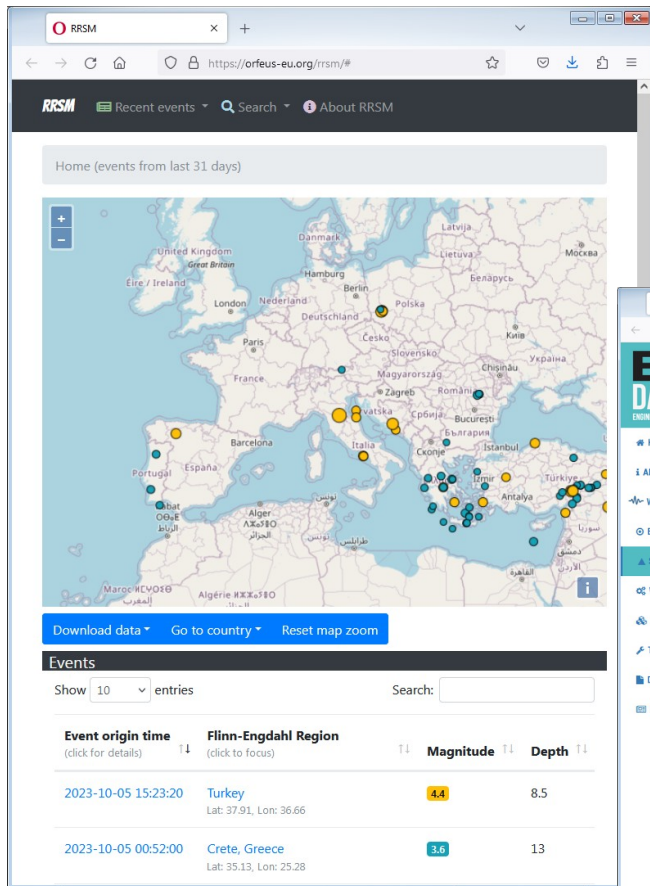
Italy M=6.5, 30 **October** 2016

New Zealand M=7.8, 13 **November** 2016

EGU 2017 talk by
Kolínský et al.
dataset: AlpArray

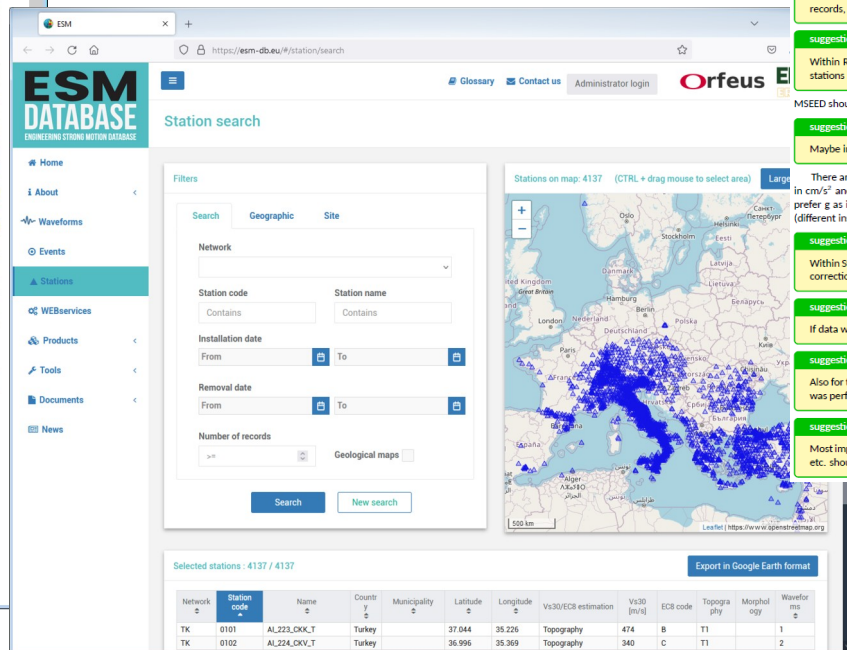
UAG 2023 report – suggestions – RRSM + ESM

recommendations for RRSM and ESM – webpage, functions



Home (events from last 31 days)

Event origin time (click for details)	Flinn-Engdahl Region (click to focus)	Magnitude	Depth
2023-10-05 15:23:20	Turkey Lat: 37.91, Lon: 36.66	4.4	8.5
2023-10-05 00:52:00	Crete, Greece Lat: 35.13, Lon: 25.28	3.6	13



Station search

Filters

Search Geographic Site

Network

Station code

Station name

Installation date

Removal date

Number of records

Geological maps

Search New search

Selected stations: 4137 / 4137

Export in Google Earth format

Network	Station code	Name	Country	Municipality	Latitude	Longitude	Vs30/EC8 estimation	Vs30 [m/s]	EC8 code	Topography	Morphology	Waveform
TK	0101	AL223_OKK_T	Turkey		37.644	35.526	Topography	474	B	T1		1
TK	0102	AL224_OKK_T	Turkey		36.996	35.369	Topography	340	C	T1		2

suggestion #8

What could be beneficial for users is to compare Pseudo-Spectral Accelerations (when you click on the Network/Station/Location) with selected GMPEs from literature.

suggestion #9

Also, it would be good to have the station information in terms of local site condition information, V₃₀, site frequency, geology and also to provide the information how these parameters were estimated (geophysical measurements, correlation, etc.) similar to the ESM portal. When you click on the event table with available records, it lists all important information about local site conditions.

suggestion #10

Within RRSM portal (<http://orfeus-eu.org/rrsm/#>) it would be good to have an option to show SM stations similar to ESM (<https://esm-db.eu/#/station/search>).

MSEED should be preferable for raw data, however, we need to think about all users.

suggestion #11

Maybe in the SM database, the ASCII format could be available together with MSEED.

There are different units used through the SM database with different countries and instruments, in g, in m/s², in cm/s² and unification is preferable. For scientific purposes, cm/s² is fine, but SM users and engineers usually prefer g as it is in seismic codes. In particular, when different instruments are set, units are in different formats (different instrument parameters) and users can be potentially mistaken when they are using a set of data.

suggestion #12

Within Station information listed, it would be good to list instrument type and have information if instrument correction was applied.

suggestion #13

If data were processed and listed together with raw data, it would be beneficial for users to show both.

suggestion #14

Also for the analysis (if it was performed), it would be good to include information on how the noise analysis was performed (with literature citation maybe), which filters were used (similar to the ESM portal).

suggestion #15

Most importantly, when a waveform is downloaded, the final file structure, header, time, acceleration values, etc. should be unified among different countries.

UAG 2023 report – suggestions - StationXML

suggestion #2

The UAG recommends ORFEUS/EIDA to promote a uniform way to produce StationXML, so that inexperienced users can be advised on the techniques and tools available to handle StationXML files.

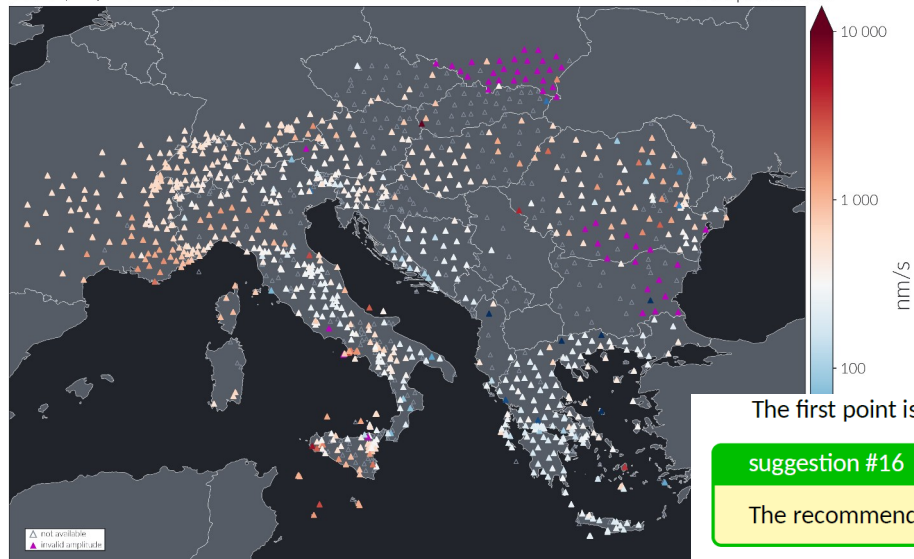
suggestion #3

In addition, ORFEUS could organize webinar or workshop sessions devoted to the management of StationXML, where station operators can share tools and discuss their direct experiences. The AdriaArray project clearly shows that training in this direction is needed.

UAG 2023 report – suggestions – metadata handling

5.0 s | Z | 2023-03-03

noise levels
95th percentile



inconsistency in handling the StationXML files by different version of SeisComP, ObsPy, (not)following the FDSN StationXML schema

The first point is that different EIDA nodes treat the metadata differently.

suggestion #16

The recommendation would be to use the same version of SeisComP at all EIDA nodes.

However, we need to note that the error was only revealed right thanks to the fact that the versions are different, and hence the different behavior could have been tracked.

suggestion #17

The more important recommendation would hence be, that the same StationXML scheme is followed by all the software developers (SeisComp, ObsPy, ...).

There is only one valid StationXML scheme, which is the one by FDSN <https://docs.fdsn.org/projects/stationxml/en/latest/>.

StationXML documentation

```
<FDSNStationXML schemaVersion="1.2">
  <Source>SeisCompP</Source>
  <Sender>GFZ</Sender>
  <Created>2023-04-02T17:49:15.915266Z</Created>
  <Network code="Z3" startDate="2015-01-01T00:00:00Z" endDate="2022-12-31T00:00:00Z" restrictedStatus="open">
    <Description>
      AlpArray Seismic Network (AASN) temporary component
    </Description>
    <Identifier type="DOI">10.12686/alparray/z3_2015</Identifier>
  </Network code="A412A" startDate="2017-06-15T16:45:38Z" endDate="2018-02-12T09:54:14Z" restrictedStatus="open">
    <Comment id="1">
```

revision of 1.1 ==> 1.2 by
Philip Crotwell
Robert Casey
Jonathan Schaeffer
Petr Kolínský
accepted in 2022 by FDSN

StationXML



v1.2

Search docs

Contents

1. Overview

2. StationXML Reference

2.1. <FDSNStationXML> required

2.2. <Network> required

2.2.1. <Description>

2.2.2. <Identifier>

2.2.3. <Comment>

2.2.4. <DataAvailability>

2.2.5. <Operator>

2.2.6. <TotalNumberStations>

2.2.7. <SelectedNumberStations>

2.3. <Station>

2.4. <Channel>

2.5. <Response>

3. Specifying and Using Response Information

4. StationXML Tools

5. Appendices

Specification home

FDSN home

2.2. <Network> required

Warning: Future Change

endDate: This attribute should not be used if it is in the future.

Warning: Future Change

endDate: This attribute is likely to require timezone of Z.

Warning: Future Change

startDate: This attribute is likely to require timezone of Z.

The Network container. All station metadata for this network is contained within this element. A Description element may be included with the official network name and other descriptive information. An Identifier element may be included to designate a persistent identifier (e.g. DOI) to use for citation. A Comment element may be included for additional comments.

An active Network should not use the endDate attribute. Unlike SEED, do not use an endDate in the distant future to mean active.

Example: <Network code="XX" startDate="2016-01-27T13:00:00Z" />

Attributes of <Network>:

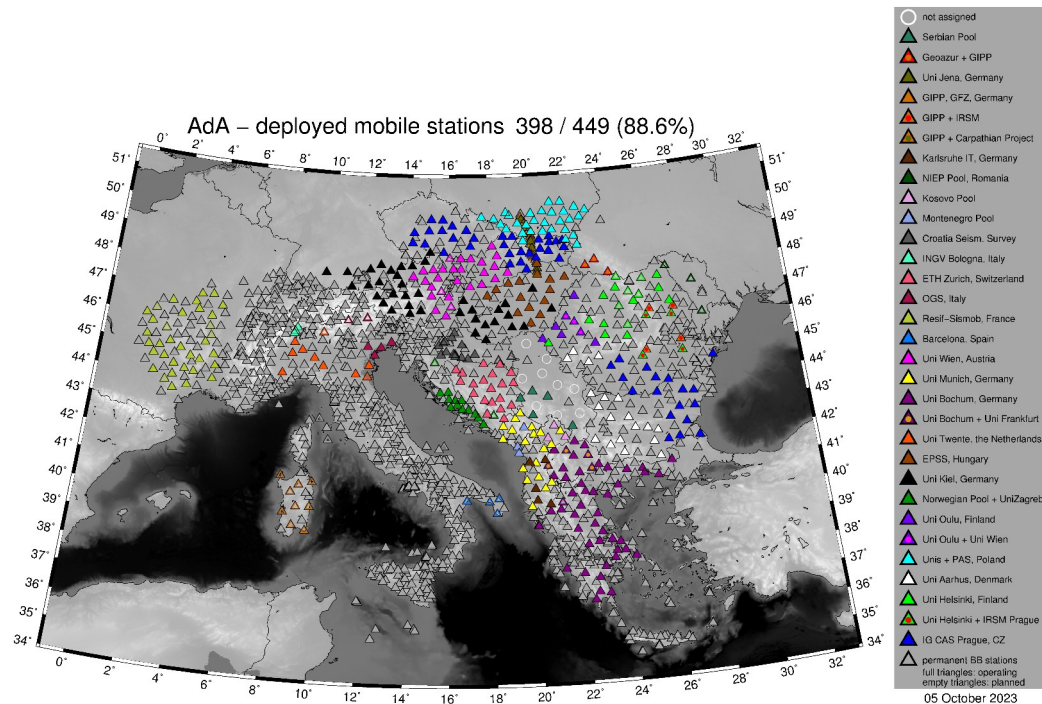
attribute	type	required	description	example
alternateCode	string	no	A code used for display or association	alternateCode="GSN"
code	string	yes	Name of Network	code="XX"
endDate	dateTime	no	End date of network. Do not use if still active, endDate should not be in the future.	endDate="2018-01-27T00:00:00Z"
historicalCode	string	no	A previously used code if different from the current code	historicalCode="XX"
restrictedStatus	RestrictedStatusType	no	One of: "open", "closed", "partial"	restrictedStatus="open"
sourceID	anyURI	no	A data source identifier in URI form. It is recommended that this follow the FDSN Source Identifiers specification, http://docs.fdsn.org/projects/source-identifiers	sourceID="FDSN:XX"
startDate	dateTime	no	Start date of network	startDate="2016-07-01T00:00:00Z"

Sub Elements of <Network>

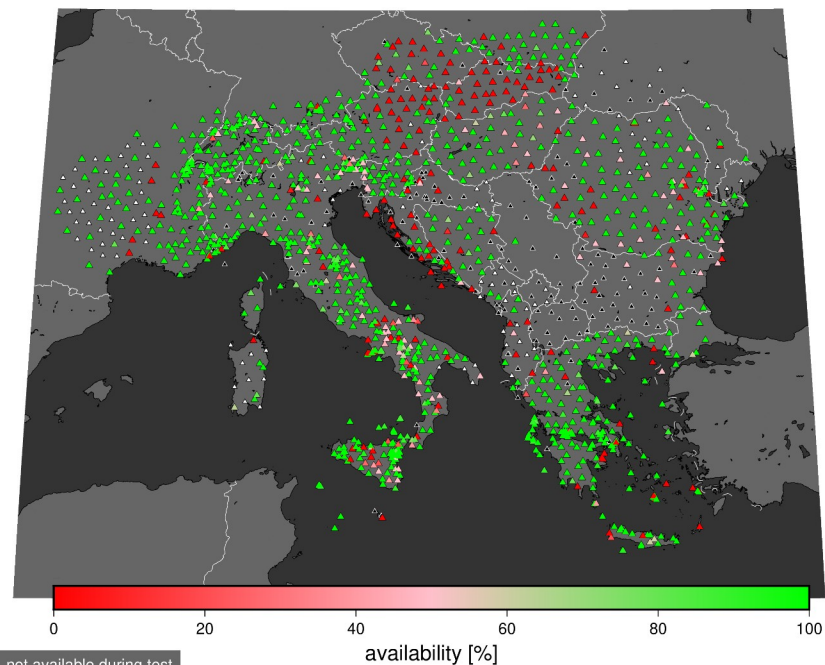
element	type	number
<Description>	string	optional

UAG and AdriaArray

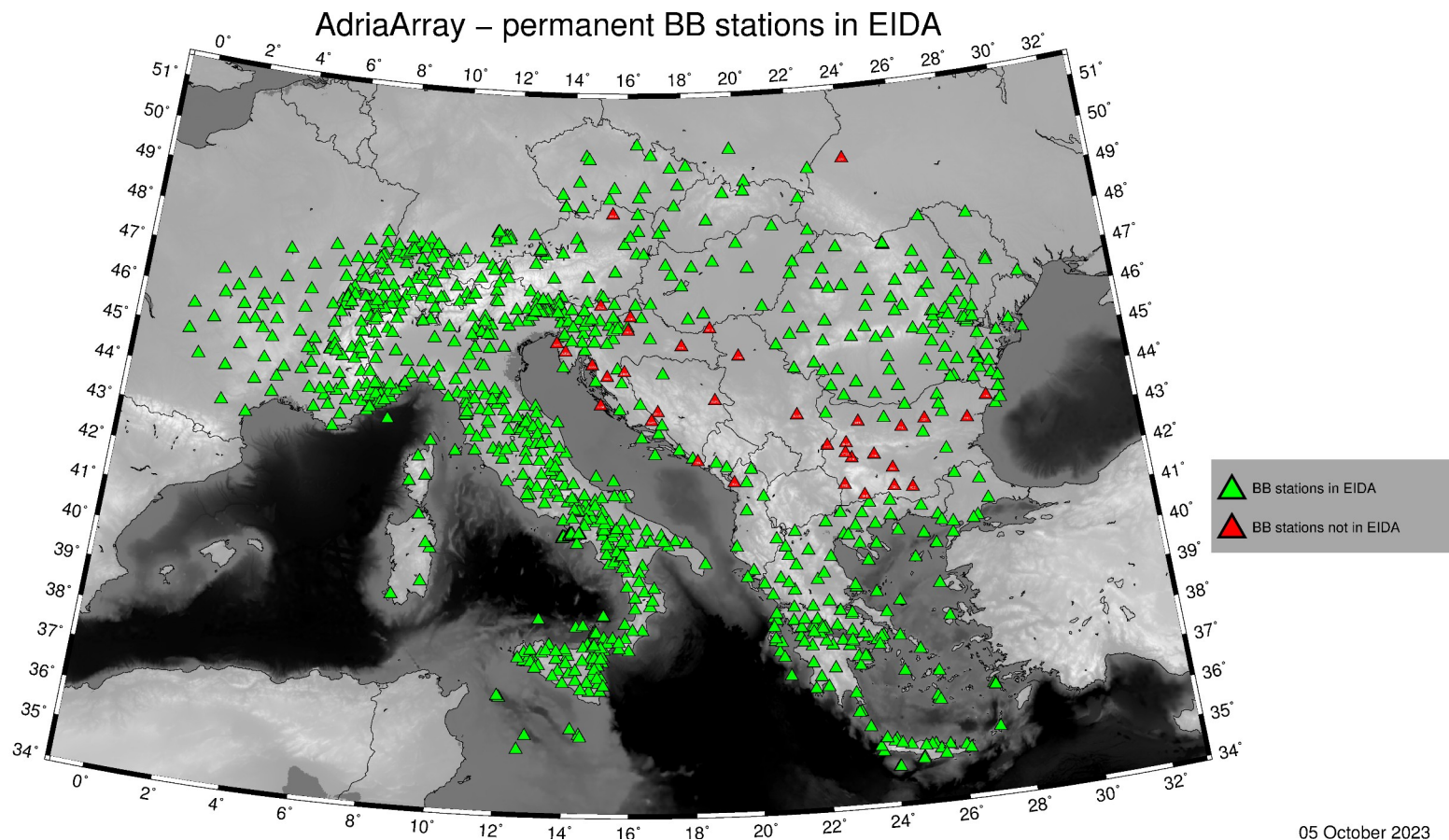
retrievability tests
by Johannes Stampa et al.
for AdriaArray



440 permanent stations



short-term vision – all permanent stations in EIDA



short-term vision – online survey

The need for a user survey was already proposed in the last UAG 2021 report, and also in the strong-motion-UAG 2020 report. The agreement updated in February 2023 at the ExeCom + UAG meeting is for the UAG to **define broader topics for the survey** and then for these topics to be translated into questions by the ExeCom and possibly other ORFEUS bodies. ORFEUS then disseminates them through its mailing lists or the newly developed ORFEUS forum. The **purpose** of the survey is to find out what **user's experiences are of ORFEUS's online services, websites, workshops and other activities**.

short-term vision – sharing the Q test results



Gidera Gröschl¹, Petr Kolínský¹, Florian Fuchs¹, Götz Bokelmann¹ and the AlpArray Working Group²

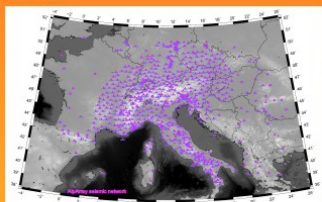
AlpArray 2016 & 2017: Data quality analysis as part of surface wave investigation

¹Department of Meteorology and Geophysics, University of Vienna, Austria, ²www.alparray.ethz.ch



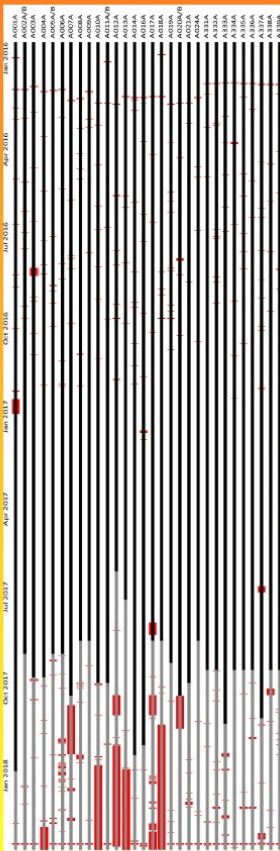
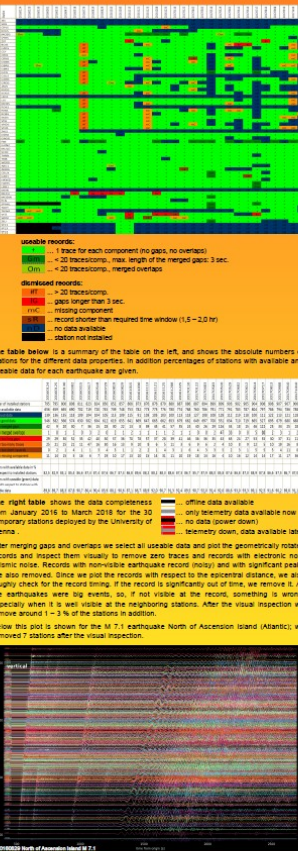
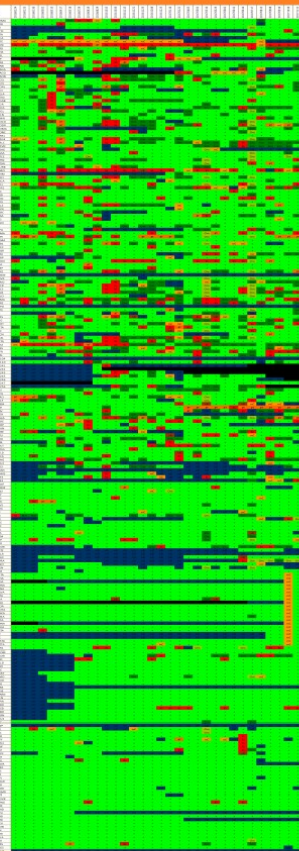
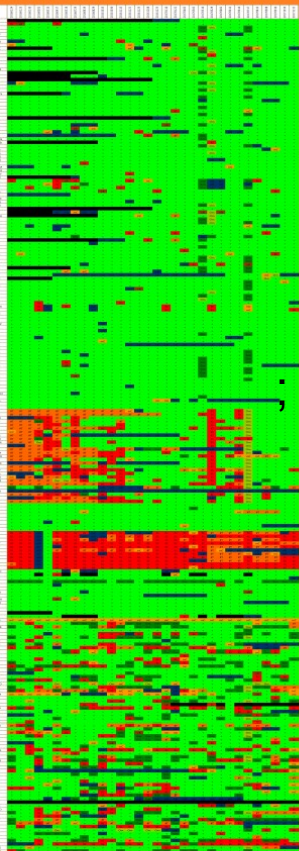
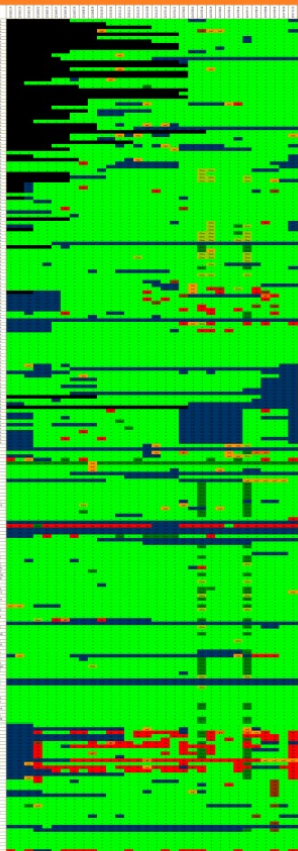
European Geosciences Union General Assembly, Vienna, Austria, 8-13 April 2018

The AlpArray research project allows us to investigate the structure and evolution of the lithosphere beneath the entire Alpine area. The project includes around 250 temporary seismic stations, which have been deployed in the period from spring 2016 to summer 2017. Together with the included permanent stations from all over Europe, there are around 500 stations available in total.



For the surface wave analysis, we are interested in two-hour time windows for each earthquake. For different reasons we cannot use the data of all stations for the data processing. These are the records that are not available for the given day, components are missing, there are too many gaps or overlaps, the traces are shifted in time, the records are too noisy or have no seismic signal. We selected 32 earthquakes spanning 2016 and 2017 and investigate the quality of the data before the surface wave analysis.

The following table contains information about availability and quality of the data for 918 stations and respective time windows for each earthquake.



vision for next years

UAG focused mainly on technical issues since its start in 2018

It is a time for broadening the scope towards the scientific targets

Kick-off meeting 2018:

The role of the UAG is to ensure **awareness** of the ORFEUS ExeCom of the main scientific, **present and future challenges** in Seismology and Earth Sciences, and to provide feedback on how ORFEUS is serving its Community to face them.

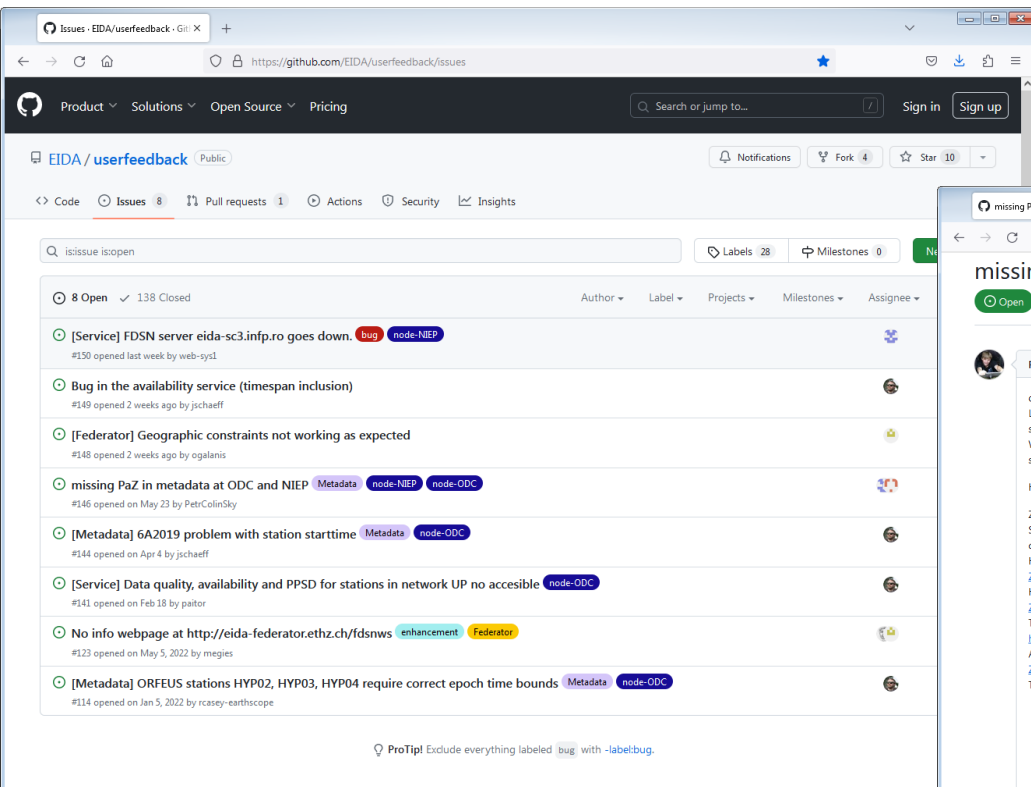
could we maybe have an EPOS Seismology UAG?

technical issues (ORFEUS) vs. (meta)data mistakes (station operators)

long-term wish: including smaller local networks in EIDA

EIDA issue tracker

<https://github.com/EIDA/userfeedback/issues>



This screenshot shows the GitHub interface for the 'EIDA/userfeedback' repository. The top navigation bar includes links for Product, Solutions, Open Source, and Pricing, along with a search bar and sign-in options. The repository name 'EIDA / userfeedback' is displayed, along with statistics for notifications, forks (4), and stars (10). The 'Issues' tab is selected, showing a list of 8 open issues. The first issue, '[Service] FDSN server eida-sc3.infp.ro goes down.', is highlighted. Other visible issues include a bug in the availability service, geographic constraints not working, and missing PaZ in metadata at ODC and NIEP. A 'ProTip!' at the bottom suggests excluding issues labeled 'bug'.

Issues · EIDA/userfeedback · Gill X

Product Solutions Open Source Pricing

Search or jump to...

Sign in Sign up

EIDA / userfeedback Public

Notifications Fork 4 Star 10

< Code Issues 8 Pull requests 1 Actions Security Insights

is:issue is:open

Labels 28 Milestones 0

8 Open 138 Closed

Author Label Projects Milestones Assignee

[Service] FDSN server eida-sc3.infp.ro goes down. bug node-NIEP

#150 opened last week by web-sys1

Bug in the availability service (timespan inclusion)

#149 opened 2 weeks ago by jschaeff

[Federator] Geographic constraints not working as expected

#148 opened 2 weeks ago by ogalanis

missing PaZ in metadata at ODC and NIEP Metadata node-NIEP node-ODC

#146 opened on May 23 by PetrColinSky

[Metadata] 6A2019 problem with station starttime Metadata node-ODC

#144 opened on Apr 4 by jschaeff

[Service] Data quality, availability and PPSD for stations in network UP no accessible node-ODC

#141 opened on Feb 18 by paitor

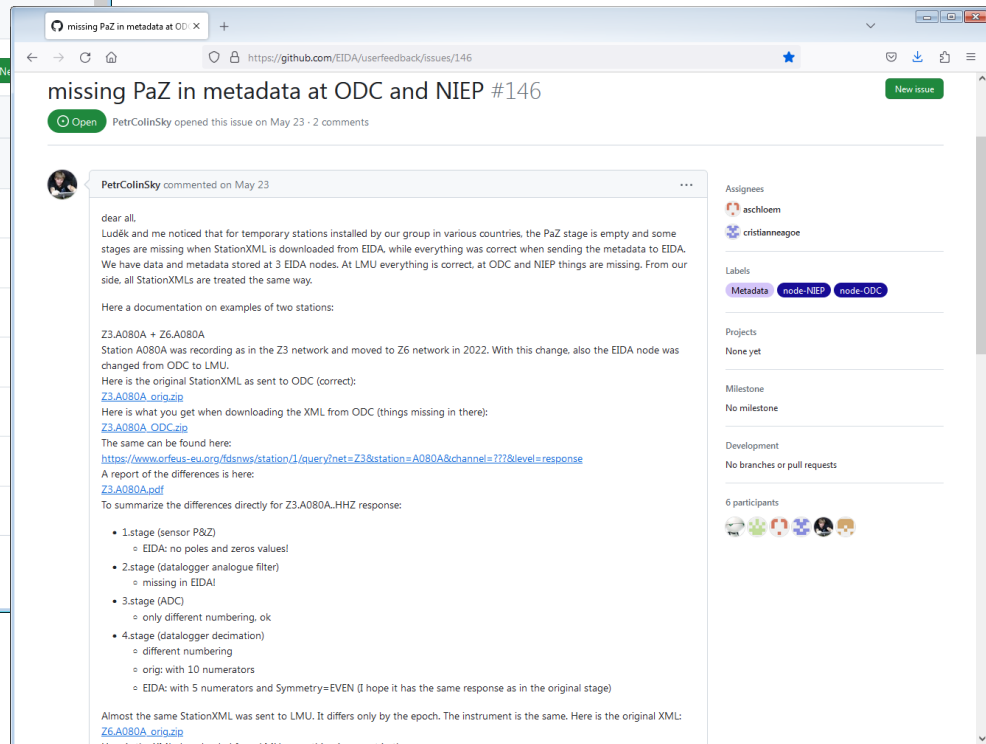
No info webpage at <http://eida-federator.ethz.ch/fdsnws> enhancement Federator

#123 opened on May 5, 2022 by megies

[Metadata] ORFEUS stations HYP02, HYP03, HYP04 require correct epoch time bounds Metadata node-ODC

#114 opened on Jan 5, 2022 by rcasey-earthscope

ProTip! Exclude everything labeled bug with -label:bug.



This screenshot shows the details of issue #146, 'missing PaZ in metadata at ODC and NIEP'. The issue is marked as 'Open' and was created by PetrColinSky on May 23. The issue description explains that for temporary stations, the PaZ stage is empty and some stages are missing when StationXML is downloaded from EIDA. It mentions that data and metadata are stored at 3 EIDA nodes (ODC, NIEP, and LMU) and that ODC and NIEP are missing. A comment by PetrColinSky from May 23 provides a detailed explanation and a list of examples of missing data for two stations: Z3.A080A and Z6.A080A. The comment includes links to the original StationXML files and a report of the differences. The right sidebar shows the issue's metadata, including assignees (aschloem, cristianneage), labels (Metadata, node-NIEP, node-ODC), projects (None yet), milestones (None), and participants (6).

missing PaZ in metadata at ODC and NIEP #146

Open PetrColinSky opened this issue on May 23 · 2 comments

PetrColinSky commented on May 23

dear all,

Luděk and me noticed that for temporary stations installed by our group in various countries, the PaZ stage is empty and some stages are missing when StationXML is downloaded from EIDA, while everything was correct when sending the metadata to EIDA. We have data and metadata stored at 3 EIDA nodes. At LMU everything is correct, at ODC and NIEP things are missing. From our side, all StationXMLs are treated the same way.

Here is a documentation on examples of two stations:

Z3.A080A + Z6.A080A

Station A080A was recording as in the Z3 network and moved to Z6 network in 2022. With this change, also the EIDA node was changed from ODC to LMU.

Here is the original StationXML as sent to ODC (correct):

[Z3.A080A_orig.zip](#)

Here is what you get when downloading the XML from ODC (things missing in there):

[Z3.A080A_ODC.zip](#)

The same can be found here:

<https://www.orfeus-eu.org/fdsnws/station/?query=net=Z3&station=A080A&channel=?&level=response>

A report of the differences is here:

[Z3.A080A.pdf](#)

To summarize the differences directly for Z3.A080A.HHZ response:

- 1.stage (sensor P&Z)
 - EIDA: no poles and zeros values!
- 2.stage (datalogger analogue filter)
 - missing in EIDA!
- 3.stage (ADC)
 - only different numbering, ok
- 4.stage (datalogger decimation)
 - different numbering
 - orig: with 10 numerators
 - EIDA: with 5 numerators and Symmetry=EVEN (I hope it has the same response as in the original stage)

Almost the same StationXML was sent to LMU. It differs only by the epoch. The instrument is the same. Here is the original XML:

[Z6.A080A_orig.zip](#)

Assignees

aschloem cristianneage

Labels

Metadata node-NIEP node-ODC

Projects

None yet

Milestone

No milestone

Development

No branches or pull requests

6 participants

aschloem cristianneage

slide from the 2019 Grenoble SM feedback



WE WANT YOU!

Please take a few minutes here or at home to
give us your own feedback based on your needs!

Reinforce/disagree with a point made
or bring up a new one!

We will process, collate & deliver to ORFEUS
to help improve!

petr.kolinsky@ig.cas.cz

2023 EPOS Seismology Workshop, 9 – 11 October, Podgorica, Montenegro



INSTITUTE OF GEOPHYSICS
OF THE CZECH ACADEMY OF SCIENCES



EPOS
EUROPEAN PLATE OBSERVING SYSTEM

Orfeus